Mohammed Khalid Shaik

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AI Software Engineer with 5+ years of experience building production AI, data, and regulated infrastructure systems where reliability, security, and auditability are mandatory. Specialized in **multi-agent orchestration**, **sandboxing**, **LLM observability**, and **GxP clinical data platforms**. Shipped agent security and monitoring for Sentinel, an AI coding-agent platform for drug research, and led clinical data infrastructure for **Takeda**, **Crinetics**, and **Jazz**, including validated AWS disaster recovery. Proven track record turning GenAI prototypes into secure, production-grade platforms for life-sciences teams.

SKILLS

Languages: Python, TypeScript/JavaScript, SQL

AI / LLM & Agents: Multi-agent orchestration, Docker sandboxes, LiteLLM, Claude, OpenAI, Bedrock, RAG, Haystack, LlamaIndex, MCP, agent evals and tooling

Backend & APIs: FastAPI, SQLAlchemy, REST APIs, Pydantic AI, async Python

Data & Workflow: PostgreSQL, Redis, Pandas, Pydantic, Airflow, BigQuery, Kafka, Snowflake, pgvector, ETL/ELT

Cloud, DevOps & Compliance: AWS S3, EBS, FSx, Lambda, EC2, RDS; Terraform, Kubernetes, Docker, CI/CD, GxP environments, GitHub Actions

Frontend & Testing: React, pytest, Locust, integration/E2E testing, AST static analysis

PROFESSIONAL EXPERIENCE

Software Engineer — Integrated Analytic Solutions

Clients: **Takeda**, **Crinetics**, **SOBI**, **Jazz**, **Neurocrine** | *Remote* | *May 2023 – Present*

- **Agent Sandboxing:** Restricted tool calls by coding agents on the Sentinel drug-research platform to explicitly allowed resources using least-privilege macOS Seatbelt sandboxes as agent guardrails.
- **Multi-Agent Orchestration:** Built a multi-agent patching system where sandboxed workers write fixes, and a read-only verifier must pass tests before merging, reducing reopened fixes and speeding up patch releases by 30% verses of unorchestrated agents.
- **Agent Observability:** Extended AI agent observability to Codex and OpenCode, tracking token spend and blocked sessions in the agent monitor.
- **FDA Submission Compiler:** Reduced clinical-study document preparation from 15 minutes to under 3 minutes per file for a service with 10,000 active users by building a FastAPI service that combines RTF outputs into submission PDFs while isolating user sessions.
- **Regulated Clinical Pipeline:** Delivered clinical-trial data from external vendors to Crinetics' biostatistics and clinical teams by building a regulated pipeline that processes 100-file bursts in under 2 minutes, with immutable GxP audit records.
- **Disaster Recovery:** Owned AWS disaster recovery for a GxP-validated R&D SaaS platform serving 10,000 users, meeting a 2-hour RTO and 4-hour RPO with a Terraform-built standby environment and Lambda automation that attaches backup EBS volumes at failover.
- **Kubernetes Reliability:** Hardened the Kubernetes engine serving 3,000 statistical programmers, delivering a SAS workspace in minutes instead of hours and fixing race conditions that killed workspaces mid-setup or left failed ones running.
- **GxP Data Migration:** Migrated SOBI's GxP data from AWS FSx and OneDrive to a single Windows server, cutting storage costs by $120K annually, verified by per-file hash checks, and access controls preserved.

Software Engineer — GameStop

Remote | *Sept 2022 – May 2023*

- **API Performance:** Reduced tested API latency from 2 seconds to 30 milliseconds across 14 FastAPI routers by moving blocking database and LDAP calls off the event loop into a thread pool.
- **Enterprise RAG:** Built an internal RAG chatbot over company pricing and promotions documents, using Haystack for embedding and retrieval on a pinecone vector store

- **Compliance ETL:** Enabled compliance tracking for the 90-day legal limit on offer runs by building Airflow ETL pipelines that load Quicklizard and Zendesk API data into BigQuery.
- **Process Automation:** Automated promotion approvals with Python, SQL Views, and REST APIs, cutting monthly review effort by 80%.

Software Engineer — M&G

India | Jan 2021 – Aug 2022

- **Identity & Access:** Kept access aligned with each user's current role for 100+ users across 15 teams by running onboarding, role changes, and offboarding in on-premises Active Directory, with Linux server access tied to AD accounts.
- **Self-Healing Infrastructure:** Automatically resolved half of incident tickets affecting a fleet of 150 Ubuntu servers by building monitoring and self-healing automation with systemd services and cron jobs.

PROJECTS

- **Agent Session Monitor:** Built Pigeon, an open-source observer that flags blocked Claude Code, Codex, and OpenCode sessions in one live view, so stalled agents get unblocked fast. github.com/pdwytr/pigeon
- **QuickXorHash CLI:** Enabled OneDrive file-hash verification during production migrations by building a Go CLI that computes Microsoft QuickXorHash. github.com/pdwytr/quickxorhash

EDUCATION

University of Texas at Arlington, *Master of Science in Data Science* — Arlington, TX | Dec 2023

Jawaharlal Nehru Technological University, *Bachelor's in Technology* — India | Aug 2020